Process & Decision Documentation

Project/Assignment Decisions

When working on this assignment I used Cursor + ChatGPT for the code. My first prompt broke the entire game when I asked it to replace the player sprite with one I found off the internet. But I realized because the format was different (instead of all sprites being in one png, they were in separate pngs. So I had to figure out a prompt that would help it consider this issue.

Role-Based Process Evidence

Changed player, enemy, and fire sprites

Name: Rose

Goal of Work Session: To change the sprites of the player, enemy, and fire
https://github.com/iiRosie1/iiRosie1_sidequest_w6/commit/a295357f988b7ccfebb54341aba7661457d4c9a8

Edited background

Name: Rose

Goal of Work Session: Changed the background to something else

https://github.com/iiRosie1/iiRosie1_sidequest_w6/commit/7a7d4e402267449ef98627e76cbc1522c2b652a7

Added Sound Effects

Name: Rose

Goal of Work Session: To add sound effects for user feedback
https://github.com/iiRosie1/iiRosie1_sidequest_w6/commit/5a213eaeca3e8ddd57f683f754ed1d6c52033a79

GenAI Documentation

If GenAI was used (keep each response as brief as possible):

Date Used: March 8, 2026

Tool Disclosure: Cursor + ChatGPT 5.2

Purpose of Use: To code and debug the sidequest

Summary of Interaction: Helped me add change the sprites, background, and add music to the game

Human Decision Point(s): Sometimes the changes Cursor makes break the game, so I have to go back and revert, then brainstorm ways to prompt better so it knows exactly what I want to achieve.

Integrity & Verification Note: I verified the code by testing the result after every change and lots of playtesting.

Scope of GenAI Use: AI helped me by giving me code I needed to execute my ideas. No AI was used to write the process and decision document.

Limitations or Misfires: Have to be very precise in prompting, or else it could change things you didn't want it to change.

Summary of Process (Human + Tool)

- Added new sprites to assets folder
- Used Cursor to make the changes to sprites and animations
- Added music and sound effects

Decision Points & Trade-offs

I wanted Cursor to change the sprites without me having to figure out how to get it working. But the change it made broke my game. So I started by manually adding in my sprites to the assets folders, then looking for the areas in the code that affect the player sprite. Then changed the file names and animation types.

Verification & Judgement

- I evaluated my changes through playtesting to see if I liked the changes or not.

Appendix

```
**Cursor**

Checking how the player uses animations and how p5play handles per-animation images:

Checking how p5play handles animation definitions and image frames:

Updating AssetLoader to load Green_Slime images and use them for player animations.

Verifying p5play's animation format for multiple images:

Using a single image for the attack animation because p5play expects paths for
multi-image animations.

Replaced the fox sprite sheet with the Green_Slime assets. Summary of changes:

**Updates in `src/AssetLoader.js`:**

1. **Loading** - Switched from `foxSpriteSheet.png` to the Green_Slime images:
  - `Idle.png`, `Run.png`, `Walk.png`, `Jump.png`, `Attack_1.png`, `Hurt.png`,
`Dead.png`

2. **`defaultSlimePlayerAnis()`** - New function that maps each animation to a single
image:
  - `idle` → Idle.png
  - `run` → Run.png
  - `walk` → Walk.png
```

- `jump` → Jump.png
- `attack` → Attack_1.png
- `hurtPose` → Hurt.png
- `death` → Dead.png

3. **Fallback sprite** – `playerImg` now uses `slimeIdle` instead of the old fox sprite sheet.

`Attack_2.png` and `Attack_3.png` are not used. To use all three attack frames, they would need to be combined into one sprite sheet, or you could try p5play’s multi-image animation support.

User

everything disappeared except for the background

Cursor

Investigating what's causing the player and other entities to disappear:

Creating a sprite sheet from the Green_Slime images at load time — p5play's addAnis expects sprite-sheet format; the `img` format may be incompatible. We'll draw the images into a grid and use the sprite-sheet layout.

Checking whether tuning uses a different key:

The problem was that p5play’s `addAnis` is built for sprite-sheet animations (row/frames), not for the `img` per-animation setup we tried. Switching to the sprite-sheet approach fixes it.

Updates in `AssetLoader.js`:

1. **`buildSlimeSpriteSheet()` added** – Combines all Green_Slime images into a 64×192 sprite sheet with 32×32 cells:
  - Row 0: Idle
  - Row 1: Run
  - Row 2: Jump
  - Row 3: Attack
  - Row 4: Walk
  - Row 5: Hurt and Death (2 frames)

2. **Animation format switched to sprite-sheet** – Player animations now use the same row/frames format as the fox sprite sheet instead of per-animation `img` values.

Everything (player, tiles, enemies, etc.) should show again because we're back to the format p5play expects. Refresh the game and confirm it looks correct.

User

Where do I change the pngs for my spirte

Cursor

You can change which PNGs the player sprite uses in **`src/AssetLoader.js`**.

Around lines 24–31, the Green_Slime images are loaded:

```24:31:src/AssetLoader.js
 const slimeBase = "assets/Green_Slime/";
 const slimeIdle = await loadImageAsync(slimeBase + "Idle.png");
 const slimeRun = await loadImageAsync(slimeBase + "Run.png");
 const slimeWalk = await loadImageAsync(slimeBase + "Walk.png");
 const slimeJump = await loadImageAsync(slimeBase + "Jump.png");
 const slimeAttack1 = await loadImageAsync(slimeBase + "Attack_1.png");
 const slimeHurt = await loadImageAsync(slimeBase + "Hurt.png");
 const slimeDead = await loadImageAsync(slimeBase + "Dead.png");
```

Change the folder or filenames there (e.g. `slimeBase` or `"Idle.png"`) to point to your new PNGs.
Other sprites are loaded in the same file, including `boarSpriteSheet.png`, `leafSpriteSheet.png`, and `fireSpriteSheet.png` (around lines 41-43).

User

The animation for jump and move left and right aren't loading for the player sprite

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Cursor

Noticing only `Pink_Monster_Idle_4.png` is loaded as the sprite sheet; it contains only the idle row. Run and jump frames are in separate files (`Pink_Monster_Run_6.png`, `Pink_Monster_Jump_8.png`). Combining those into a sprite sheet so run and jump load correctly.

The problem was that only `Pink_Monster_Idle_4.png` was being used as the sprite sheet. That image has only the idle row, so there were no run or jump rows.

The loader now combines all six Pink_Monster strips into one sprite sheet:

Row	Animation	Frames
0	Idle	4
1	Run	6
2	Jump	8
3	Attack	4
4	Hurt	4
5	Death	8

Idle, run, and jump should now show correctly when standing still, moving, and jumping.

User

Replace boar with relevant @assets/3 Dude_Monster sprite

Cursor

The boar enemy is now using the Dude_Monster sprites from `assets/3 Dude_Monster/`:

- **Run** → `Dude_Monster_Run_6.png` (6 frames)
- **Throw pose** → `Dude_Monster_Throw_4.png`
- **Death** → `Dude_Monster_Death_8.png` (8 frames)

These are combined into one sprite sheet and wired to the boar's run, throwPose, and death animations.

User

Replace the leaf with relevant @assets/2 Owlet_Monster sprite

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Cursor

The leaf collectible is now using `Owlet_Monster_Idle_4.png` from `assets/2 Owlet_Monster/`. Changes made:

- **AssetLoader.js**: `leafImg` now loads `Owlet_Monster_Idle_4.png`
- **TIleBuilder.js** and **Leaf.js**: Idle animation updated from 5 frames to 4 to match the new sprite sheet

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User

Make the owlet sprite half its current size

Cursor

Set `scale = 0.5` on the leaf group in both `TIleBuilder.js` and `Leaf.js`, so the Owlet collectible renders at half its previous size.

User

It didn't change size, make it smaller

Cursor

Using the group’s `scale` wasn’t affecting the sprites. `scale` is now set on each leaf sprite right after spawn in the post-spawn loop, at 0.35 (about one-third of the original size). The Owlet collectibles should appear clearly smaller now.

User

Replace the owlet with @assets/coin.png (it has 12 frames)

Cursor

The collectible is now `assets/coin.png` with its 12-frame rotation animation. Updates were made in `AssetLoader.js`, `TIleBuilder.js`, and `Leaf.js`. The 0.35 scale on each collectible sprite is unchanged, so coins will still render at the smaller size.

User

Replace the fire with @assets/slime_purple.png it has four frames in the first row. use those four

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Cursor

Replaced the fire hazard with `assets/slime_purple.png`, using the first row (row 0) with 4 frames. Updates made in `AssetLoader.js`, `TIleBuilder.js`, and `Fire.js`.

User

The slime size is 96 × 72 pixels

Cursor

Updated the slime (fire hazard) to use 96 × 72 pixel frames in `TIleBuilder.js` and `Fire.js`.

User

I meant the entire image is 96x72 but it has 12 sprites on it each.

Cursor

Adjusted the slime frame size for a 96×72 sheet with 12 sprites (4×3 grid), so each frame is 24×24. Both `TIleBuilder.js` and `Fire.js` now use these dimensions and the 4 frames in the first row.

User

Make the hitbox for the slimes smaller to fit the size of the sprite

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Cursor

Set the slime hitbox to 14×14 pixels in both `TIleBuilder.js` and `Fire.js`, so the collision area better matches the visible sprite while the animation stays at 24×24.

User

Where do i edit the background

Cursor

You can change the background in these places:

1. `data/levels.json`
- **Background color** – `level.view.background` (RGB array):
 ```json
 "background": [69, 61, 79]
 ```
- **Parallax layers** – `level.view.parallax` (images and scroll speed):
 ```json
 "parallax": [
   { "img": "assets/background_layer_1.png", "speed": 0.2 },
   { "img": "assets/background_layer_2.png", "speed": 0.4 },
   { "img": "assets/background_layer_3.png", "speed": 0.6 }
 ]
 ```
 Change `img` paths or `speed` for each layer.

2. `src/AssetLoader.js`
- Default background images when `levels.json` has no parallax (around lines 287–291):

- `bgFar`: `assets/background_layer_1.png`
- `bgMid`: `assets/background_layer_2.png`
- `bgFore`: `assets/background_layer_3.png`

3. Replace image files
- Update or replace the PNGs in `assets/` (e.g. `background_layer_1.png`, etc.) and keep the same filenames, or update their paths in `levels.json` or `AssetLoader.js`.

User

@assets/back.png @assets/front.png @assets/middle.png replace the current background with these

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Cursor

Updated `data/levels.json` so the parallax layers use your new backgrounds:

- **back.png** – far layer, speed 0.2
- **middle.png** – middle layer, speed 0.4
- **front.png** – near layer, speed 0.6

Layers are ordered back → middle → front and use the correct `.png` extensions.

User

Play @assets/time_for_adventure.mp3 in the background. Play@assets/tap.wav when space bar is pressed, play @assets/explosion.wav when a slime is touched, play @assets/jump.wav when the player jumps, play @assets/hurt.wav when the player gets hit

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Cursor

Sound setup is in place:

SoundManager.js
- Added `loadBGM()` and `playBGM()` for looping background music.

main.js
- Loads:
 - `assets/time_for_adventure.mp3` – BGM
 - `assets/tap.wav` – attack (space)
 - `assets/explosion.wav` – slime touch
 - `assets/jump.wav` – jump
 - `assets/hurt.wav` – player hit
- Starts BGM when audio is first unlocked (first click/key press).

Game.js
- `player:attacked` → `tap`
- `player:jumped` → `jump`
- `fire:touched` → `explosion`
- `player:damaged` → `hurt`

Level.js
- Emits `fire:touched` when the player overlaps a slime hazard.

Add `assets/time_for_adventure.mp3` if it isn't in the project yet; `explosion.wav`, `hurt.wav`, and `jump.wav` are already present.

User

Play @assets/coin.wav when the tiny owlet is rescued/picked up/touched

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Cursor

Configured `assets/coin.wav` to play on collectible pickup: added loading in `main.js` and wired it to the `leaf:collected` event in `Game.js`.